Multi-Task Paradigms in Eye-Tracking: Enhancing Diagnostic Social Attention Pattern Detection in Autism Spectrum Disorder via Machine Learning

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Background

Why Eye-Tracking?

- Objective, time-efficient, and precise
- A powerful tool for identifying autism biomarkers (Han et al., 2026)
- Many eye-tracking tasks show strong predictive performance in autism diagnosis

The Problem

- **No standardized guidelines** for task or feature selection (Setu, 2025)
- Findings remain task-specific and hard to generalize

Our Approach

- Identify shared eye-movement patterns **across different tasks**

→ To enhance **generalizability** and support real-world diagnostic application

Objectives

1. To evaluate and compare the contributions of **multiple social eye-tracking tasks** to autism detection across **multiple machine learning algorithms**.
2. By systematically comparing feature importance across tasks, identifying eye-movement features that serve as **universal markers for autism diagnosis**.

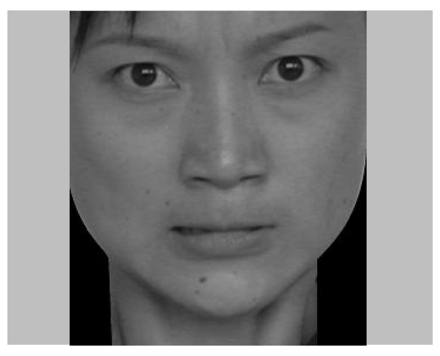
Methods

Participants

ASD group (n = 249)
♂ : ♀ = 5.9 : 4.1
Age: 5.55 ± 1.65 years

TD group (n = 306)
♂ : ♀ = 4.8 : 5.2
Age: 4.59 ± 0.98 years

Eye-tracking Tasks



① Facial Expression Recognition

② Responding Joint Attention

③ Initiating Joint Attention

Data Analysis

Features
- **Device**: Tobii TX-300 eye tracker (300 Hz)
- **52 eye-movement features**: fixation duration, visit count etc.

Modeling
- **Models**: linear (e.g. SVM) & non-linear (e.g. XGBoost) models
- **Input**:
  - Individual tasks
  - Combined tasks
- **Data split**: Train / Val / Test = 6:2:2
- **Optimization**:
  - RandomizedSearchCV / GridSearchCV
  - Cross-validation

Model Interpretation
- **SHapley Additive exPlanations (SHAP)**
- Compared feature importance **within and across tasks**

Cross-task Comparison
- Examined **separate** eye-movement patterns in each task
- Evaluated **generalizability across tasks**

Results

Model Performance

- Combined-task models achieved strong performance, with XGBoost performing best.
- FER yielded the highest accuracy, while IJA and RJA showed moderate performance.

Task	AUC	Accuracy	F1
Combined	0.83 – 0.86	0.74 – 0.81	0.72 – 0.80
FER	0.96 – 1.00	0.97 – 1.00	0.96 – 1.00
IJA	0.76 – 0.90	0.68 – 0.83	0.55 – 0.79
RJA	0.73 – 0.78	0.63 – 0.71	0.57 – 0.70

SHAP Results

① Combined-task Feature Importance

- Facial-region gaze features were the most influential predictors.

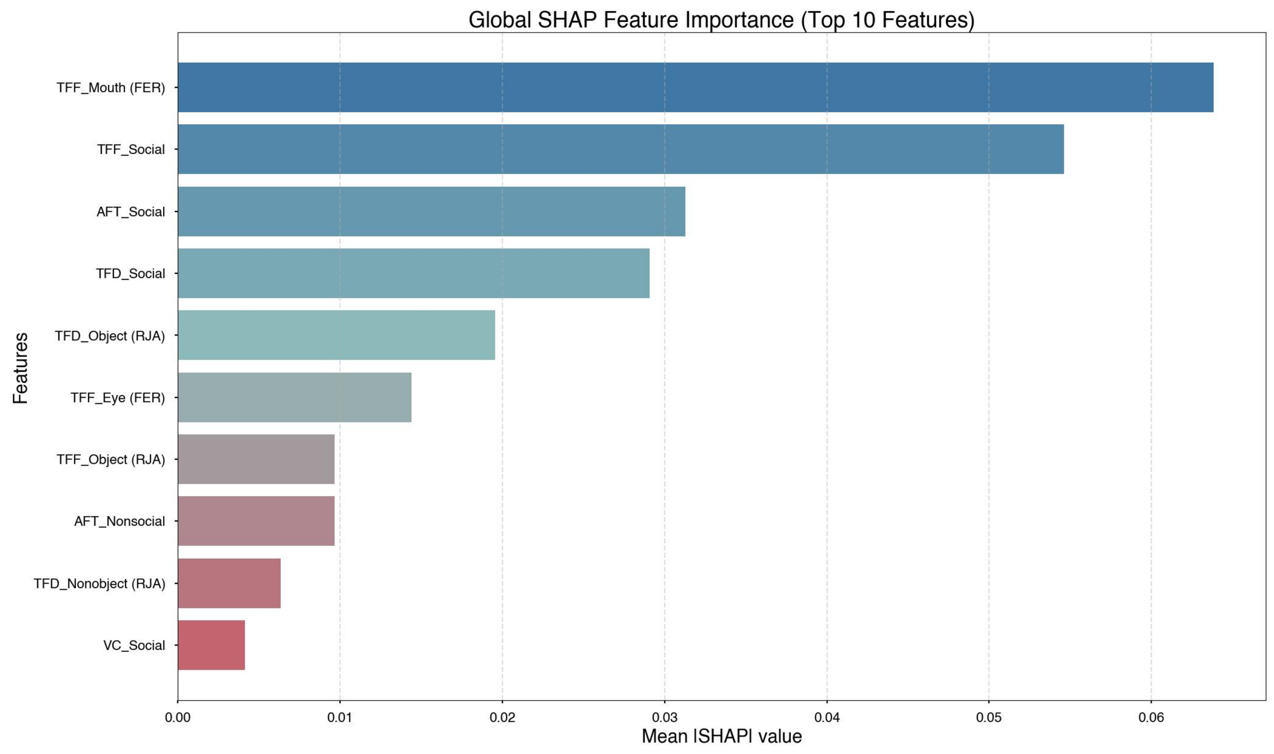


*Figure*1 SHAP (SHapley Additive exPlanations) feature importance of each eye-tracking feature for predicting ASD and TD children. The mean absolute SHAP value per feature is presented in the bar plot with larger bar plots displaying higher importance of the feature in discriminating between the two groups, and only features agreed on by at least four models are displayed. Features followed by a task abbreviation in parentheses (i.e., FER, IJA, RJA) are specific to that task; features without parentheses are shared across all three tasks. The suffix 'Social' denotes social-part/scene features, while face-part features are labelled accordingly.FER: facial emotion recognition; IJA: initiating joint attention; RJA: responding joint attention; TFF: time to first fixation; TFD: total fixation duration; AFT: average fixation time; VC: visit count.

② Cross-task Feature Patterns

- Social-region (face-related) features remained consistently important across tasks.

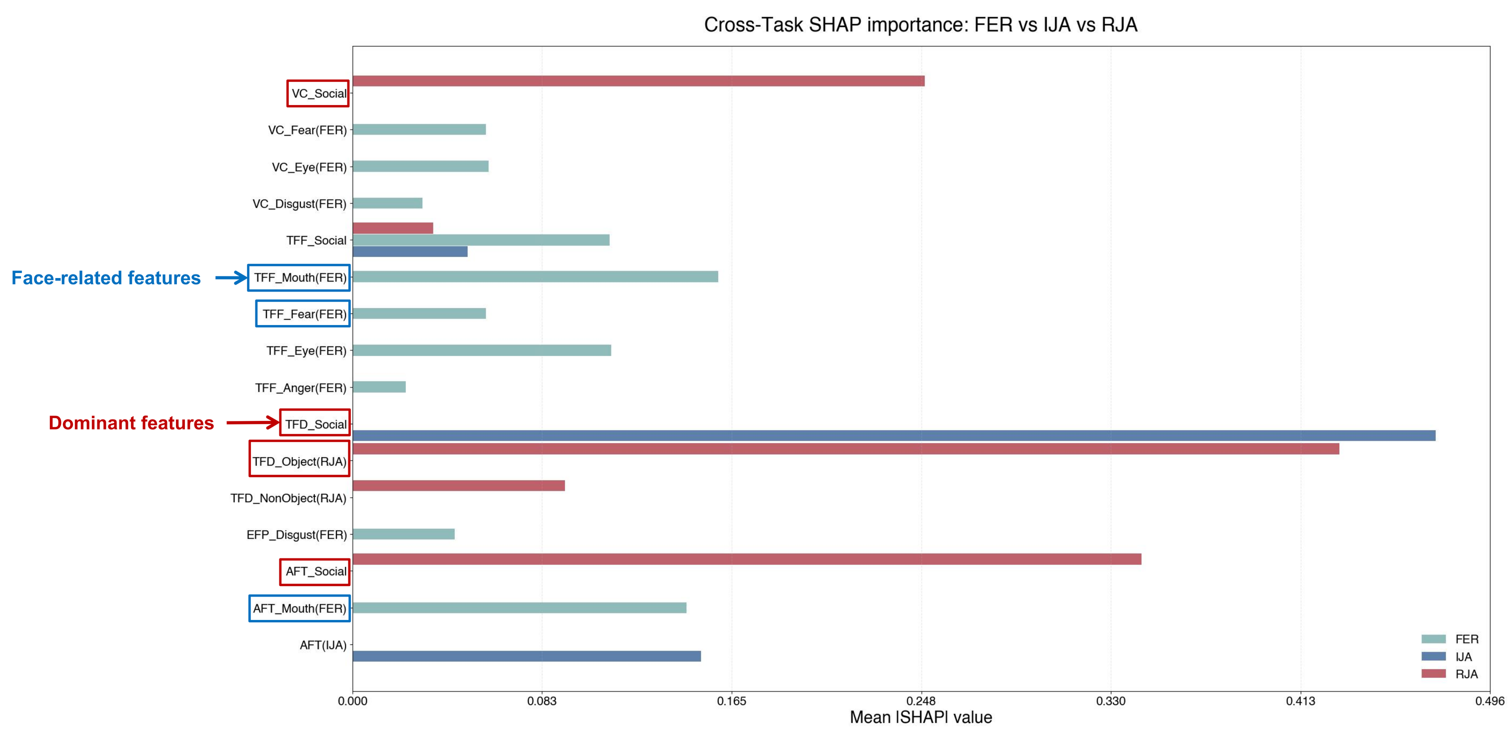


*Figure*2 SHAP feature importance for predicting ASD and TD children in different tasks. Features appearing in only one or two task bars indicate that the variable mainly drives prediction within those specific tasks. The suffix 'Social' denotes social-part/scene features, while face-part features are labelled accordingly. FER: facial emotion recognition; IJA: initiating joint attention; RJA: responding joint attention; TFF: time to first fixation; TFD: total fixation duration; EFP: eye fixation proportion;AFT: average fixation time; VC: visit count.

Conclusion

Key Findings

- A **multi-task eye-tracking + machine learning framework** effectively identifies diagnostic patterns of social attention in autism
- The **facial expression recognition task** showed the strongest discriminative power
- **Facial-related gaze features** contributed significantly across tasks → suggesting attention to facial cues is a **core cross-task dimension** of social attention atypicalities in ASD

Clinical Implications

- The framework offers a pathway toward more **robust and generalizable autism biomarkers** and stronger foundation for **real-world diagnostic application**

Reference

Setu, D. M. (2025). A systematic review of developments in eye tracking and machine learning for the early detection of autism spectrum disorder. *Journal of Technology in Behavioral Science*, 1–18. https://doi.org/10.1007/s41347-025-00542-x

Han, W., Yang, X., Li, X., Wang, J., Liu, J., & Pang, W. (2026). Machine learning–based diagnosis of autism spectrum disorder in children and adolescents using eye-tracking data: A systematic review and meta-analysis. *International Journal of Medical Informatics*, *208*, 106235. https://doi.org/10.1016/j.ijmedinf.2025.106235